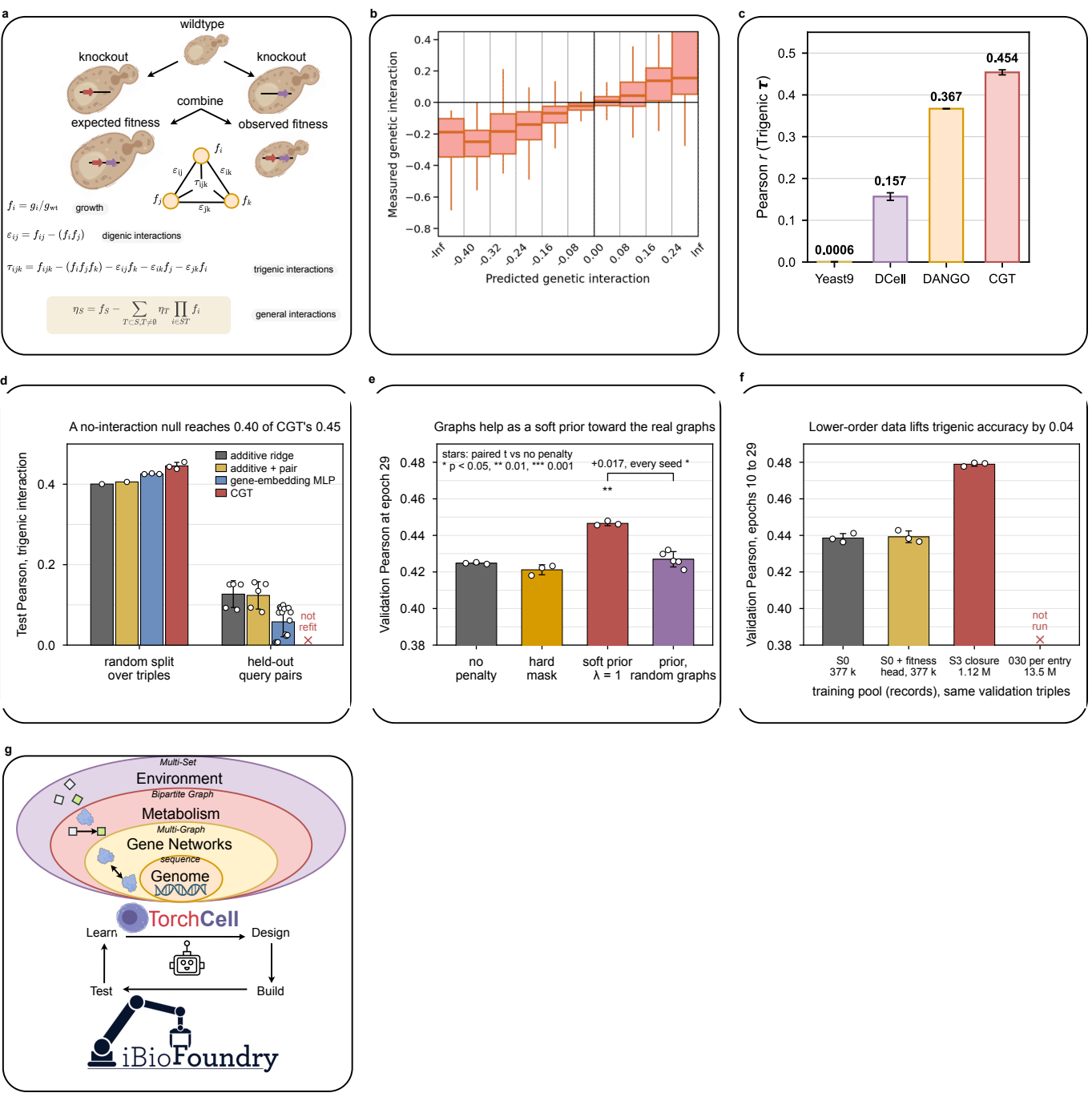

a
wildtype
knockout
knockout
combine
expected fitness
observed fitness
growth
digenic interactions
trigenic interactions
general interactions
b
Measured genetic interaction
Predicted genetic interaction
c
Pearson r (Trigenic τ)
0.0006
0.157
0.367
0.454
Yeast9
DCell
DANGO
CGT
d
A no-interaction null reaches 0.40 of CGT's 0.45
Test Pearson, trigenic interaction
additive ridge
additive + pair
gene-embedding MLP
CGT
not refit
random split over triples
held-out query pairs
e
Graphs help as a soft prior toward the real graphs
Validation Pearson at epoch 29
stars: paired t vs no penalty
* p < 0.05, ** 0.01, *** 0.001
+0.017, every seed *
**
no penalty
hard mask
soft prior λ = 1
prior, random graphs
f
Lower-order data lifts trigenic accuracy by 0.04
Validation Pearson, epochs 10 to 29
not run
S0 377 k
S0 + fitness head, 377 k
S3 closure 1.12 M
030 per entry 13.5 M
training pool (records), same validation triples
g
Multi-Set
Environment
Bipartite Graph
Metabolism
Multi-Graph
Gene Networks
sequence
Genome
TorchCell
Learn
Design
Test
Build
iBioFoundry